13 At a certain factory Christmas tree decorations are packed in boxes of 10.

The quality control manager collects a random sample of 100 boxes of decorations and records the number of decorations in each box which are damaged.

His results are displayed in **Fig. 13.1**.

Number of damaged decorations	0	1	2	3	4	5 or more
Number of boxes	19	35	28	13	5	0

Fig. 13.1

(a) Calculate

- the mean number of damaged decorations per box,
- the standard deviation of the number of damaged decorations per box. **[2]**

It is believed that the number of damaged decorations in a box of 10, X, may be modelled by a binomial distribution such that $X \sim \mathrm{B}(n, p)$.

(b) State suitable values for n and p. **[1]**

(c) Use the binomial model to complete the copy of **Fig. 13.2** in the Printed Answer Booklet, giving your answers correct to **1** decimal place. **[3]**

Number of damaged decorations	0	1	2	3	4	5 or more
Observed number of boxes	19	35	28	13	5	0
Expected number of boxes						

Fig. 13.2

(d) Explain whether the model is a good fit for these data. **[1]**

13(a)	
	mean =
	standard deviation =
13(b)	
	n =
	p =

13(c)

Number of damaged decorations	0	1	2	3	4	5 or more
Observed number of boxes	19	35	28	13	5	0
Expected number of boxes						

13(d)